

Name: AVIRAJ KUMAR RAI AVIRAJ KUMAR RAI | Reg. No: RA2511003011026

Unit 5 : Session 6 : SLO 2 | Symbolic Limits and Series

I. Questions & Answers

- $\lim_{x \rightarrow \infty} 1/x = 0$ as x grows without bound, $1/x$ shrinks to 0. In SymPy: `limit(1/x, x, oo)` $\rightarrow 0$.
- Sum of $1 + 1/2 + 1/4 + \dots$ a geometric series with first term $a = 1$ and ratio $r = 1/2$; $\text{sum} = a/(1 - r) = 1/(1 - 1/2) = 2$.
- $\lim_{x \rightarrow 0} \sin(x)/x = 1$ the classic standard limit. In SymPy: `limit(sin(x)/x, x, 0)` $\rightarrow 1$.

II. Code

- `from sympy import symbols, limit, sin, oo, summation`
- `from sympy.abc import x, n`
- `print(limit(1/x, x, oo)) # 0`
- `print(summation((1/2)**n, (n, 0, oo))) # 2`
- `print(limit(sin(x)/x, x, 0)) # 1`